

```

7 class DLL:
8     def __init__(self):
9         self.head = None
10
11     def insert_at_position(self, ur_position, data):
12         new_node = Node(data)
13
14         #insert at start
15         if ur_position == 1:
16             if self.head is not None:
17                 new_node.next = self.head
18                 self.head.prev = new_node
19                 self.head = new_node
20             return
21
22
23         # 2 or more node
24         current_node = self.head
25         position = 1
26
27         while current_node is not None and position < ur_position - 1:
28             current_node = current_node.next
29             position += 1
30
31         if current_node is None:
32             print("invalid postion")
33             return
34
35         new_node.next = current_node.next
36         if current_node.next is not None:
37             current_node.next.prev = new_node
38
39         current_node.next = new_node
40         new_node.prev = current_node
41
42     def print_list(self):
43         current_node = self.head
44         while current_node is not None:
45             print(current_node.data)
46             current_node = current_node.next

```



Search




```
58
59 def print_list(self):
60
```

```
61     current_node = self.head
62
```

```
63     while current_node is not None:
64
```

```
65         print(current_node.data , end = " <-> ")
66         current_node = current_node.next
67
```

```
68     print("None")
69
```

```
70 def search(self, key):
71
```

```
72     if self.head is None:
73
```

```
74         print("list is empty")
75         return
76
```

```
77     current_node = self.head
78
```

```
79     while current_node is not None:
80
```

```
81         if current_node.data == key:
82             print(f"key {key} found")
83             return
84
```

```
85         current_node = current_node.next
86
```

```
87     print(f"key {key} not found")
88
```

```
89 dll = DLL()
90
```

```
91 dll.insert_at_beginning(50)
92
```

```
93 dll.insert_at_beginning(40)
94
```

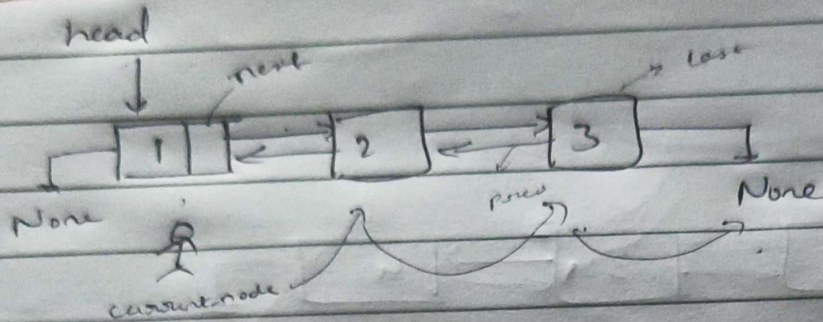
```
95 dll.insert_at_beginning(30)
96
```

```
97 dll.insert_at_beginning(20)
98
```

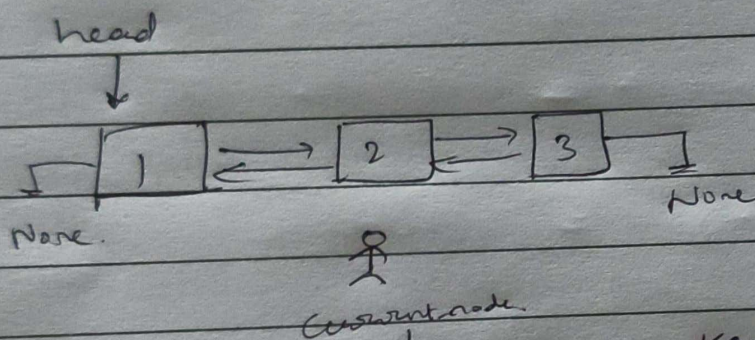
```
99 dll.insert_at_beginning(10)
100
```


13 Doubly Linked List Point & Search

* point :-



* Search :-



$\rightarrow \text{data} == \text{key}$

`print("found")`

cases :-

$\rightarrow$ Empty

$\rightarrow 1$

$\rightarrow > 1$